

Sri Lanka Institute of Information Technology



Streetify

Web-Based Transportation System

Assignment Part 01 Enhanced Entity Relationship Diagram Part A - Requirement Gathering

Group ID : 2026-Y2-S1-KU-24 (Group 24)

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System Purpose

Streetify is a centralized, cloud-hosted web-based transportation and fleet dispatching platform developed for SwiftRide Lanka (Pvt) Ltd. The primary objective is to replace disparate ride-hailing communications with an automated, low-latency matching engine that connects passengers with vetted drivers. The system automates the complete operational lifecycle: fare estimation based on GPS telemetry, real-time dispatching, multi-channel payment settlements (Card, Digital Wallet, Cash), automated platform commission accounting, customer support dispute resolutions, and administrative compliance monitoring.

User Profiles and Stakeholders

User Role	Category	Primary Responsibilities & Access Scope
Passenger	End User	Registers account, calculates upfront fare estimates, books rides, tracks assigned drivers via GPS, executes digital/cash payments, submits ratings, and lodges dispute tickets.
Driver	Service Provider	Submits commercial verification documents, toggles availability status, accepts/rejects trip requests, updates ride milestones (<i>Arrived, Started, Completed</i>), and views earnings ledgers.
Senior Driver Coordinator	Operation Staff	Verifies identity, driving licenses, revenue licenses, and vehicle fitness certificates; manages driver onboarding approvals and investigates cancellation rate anomalies.
Customer Experience Officer	Support Staff	Inspects ride telemetry, cancellation logs, and passenger claims; processes fee refunds and issues formal driver misconduct warnings.
Finance Manager	Accounting Staff	Monitors platform commission splits, approves payout reconciliations, investigates chargebacks, and exports monthly fiscal statements.
System Administrator	Governance Staff	Manages Role-Based Access Control (RBAC), audits technical system logs, monitors peak demand analytics, and enforces policy-violating account suspensions.

Functional Requirements (FR)

Module 1 : User Account & Verification Management

FR-UAM-01: The system shall allow Passengers and Drivers to register, authenticate, and reset passwords via multi-factor SMS/Email OTP verification.	FR-UAM-02: The system shall provide a document upload pipeline for drivers to submit driving licenses, revenue licenses, and vehicle inspection certificates in PDF/JPEG format.	FR-UAM-03: The system shall provide Senior Driver Coordinators an administrative portal to approve, reject (with mandatory feedback), or blacklist driver applicant profiles.
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Module 2 : Ride Booking & Driver Dispatch Engine

FR-RBM-01: The system shall compute real-time route distance, travel duration, and dynamic fare estimates upon entry of pickup and destination coordinates.	FR-RBM-02: The system shall broadcast ride requests to active drivers within an optimal municipal radius using a 30-second acceptance timeout.	FR-RBM-03: The system shall allow passengers to cancel a pending request prior to driver acceptance with zero penalty.
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Module 3 : Trip Progress & Telemetry Tracking

FR-DTM-01: The system shall enforce sequential status milestone transitions by the driver: Arrived → In Progress → Completed	FR-DTM-02: The system shall track and stream GPS telemetry and detect route deviations to recalibrate final metered fares dynamically.	FR-DTM-03: The system shall enable drivers to cancel rides with a standard penalty applied to passengers if waiting time exceeds 5 minutes (<i>Passenger No-Show</i>).
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Module 4 : Multi-Channel Payment & Ledger Accounting

FR-PRM-01: The system shall process digital transactions via external Payment Gateways (Credit/Debit Card, Digital Wallet) and record physical cash payment confirmations.	FR-PRM-02: The system shall automatically deduct company commission percentages and credit net balances to the driver's virtual wallet upon trip completion.	FR-PRM-03: The system shall generate an immutable digital PDF invoice/receipt and dispatch it via the Notification Service to the passenger's registered email.
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Module 5 : Feedback, Rating & Dispute Resolution

FR-RFM-01: The system shall prompt passengers to submit 1–5 star ratings and qualitative reviews following ride completion.	FR-RFM-02: The system shall allow passengers to lodge cancellation fee disputes within 7 days of trip termination.	FR-RFM-03: The system shall provide Customer Support Officers with investigative consoles displaying linked GPS telemetry, cancellation timestamps, and one-click refund command triggers.
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Module 6 : Administration, Security & System Control

FR-ASC-01: The system shall provide real-time dashboard analytics displaying active trips, fleet density, and demand surge heatmaps.	FR-ASC-02: The system shall allow System Administrators to suspend or permanently deactivate policy-violating accounts and immediately revoke active JWT session tokens.	FR-ASC-03: The system shall record tamper-proof audit trails for all administrative status changes and financial interventions.
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Non-Functional Requirements (NFR)

Metric	Category	Specification Standard
NFR-PERF-01	Performance	Driver dispatch matching and fare computation responses shall execute within ≤ 2.0 seconds under peak load.
NFR-SEC-01	Security	All API endpoints shall enforce HTTPS/TLS 1.3 encryption, Role-Based Access Control (RBAC), and bcrypt hashing for stored passwords.
NFR-REL-01	Availability	The platform shall maintain a 99.9% uptime availability, utilizing automated database connection pooling and transaction rollbacks.
NFR-SCAL-01	Scalability	The relational database architecture must support up to 10,000 concurrent active ride sessions without database deadlocks.
NFR-DATA-01	Data Integrity	Financial ledgers and transaction records must adhere strictly to ACID compliance to eliminate double-crediting or orphaned billing records.

Constraints and Assumptions

Constraints	Assumptions
Regulatory Compliance: Driver onboarding must strictly enforce Sri Lankan motor traffic licensing validations and passenger privacy standards.	GPS location services are enabled on passenger and driver mobile browser client devices.
Network Latency: Telemetry streaming must function on mobile web browsers operating under intermittent 3G/4G bandwidth constraints.	Standard commission rates (e.g., 15–20%) are globally configured by the platform administrator and applied uniformly unless promotional overrides exist.
Payment Settlement Boundary: External gateway interactions are restricted to tokenized API callbacks; the platform does not store raw credit/debit card numbers	The external notification service (Email/SMS gateway) delivers dispatched alerts with minimal upstream queuing delay.

Our Requirement Gathering Process

We have been conducted several Requirements gathering activities and here are the examples of that. (Evidence of the requirement gathering activities mentioned below as Screenshots)

Hi, I'm doing a software project for our software engineering module. Can you help with that? Copilot Of course! I'd be happy to help with your software engineering project. First, please tell me a little about your project: 1. What is the project title? 2. Who will use the system? 3. Is it a web application, mobile app, desktop system, or something else? 4. What does the system do? Once you give these details, I will create a Sri Lankan client company scenario with 5 different people (personas) from that company. Then you can talk to them and collect requirements just like a real client meeting. 🗨️		okay thank you very much , can i talk to Mr. kasun jayawardena Copilot Kasun Jayawardena (Senior Driver Coordinator): Hello! Nice to meet you. I work closely with our drivers and manage driver-related operations. My responsibility is to make sure drivers are registered properly, available for rides, and follow company policies. You can ask me about driver registration, trip request handling, ride acceptance, ride cancellation, driver ratings, or any challenges drivers face in daily operations. 🗨️
1.Steetify web based transport system like uber 2.passengers,drivers,system administrators 3.web application 4.user account management ,ride booking,driver trip request management,review and feedback management,payment and receipt system,admin and system control		How do drivers accept or cancel trips? Copilot Kasun Jayawardena (Senior Driver Coordinator): Sure. When a new ride request comes, available drivers receive a notification. They can see basic trip details such as pickup location and drop-off location. If a driver wants the trip, they click Accept Ride. After accepting, the system sends the driver's details to the passenger. Sometimes drivers need to cancel a trip. This may happen because of a vehicle problem, an emergency, or difficulty reaching the pickup location. When cancelling, the driver should give a reason. We monitor cancellations because too many cancellations can affect service quality. For our business, it is important that drivers do not cancel rides often after accepting them.
Great! Your project is clear. Project Name: Streetify Web-Based Transport System We are a transport service company in Sri Lanka called SwiftRide Lanka (Pvt) Ltd. We want a web system similar to Uber to help passengers book rides and help drivers manage trips. The system should also support payments, reviews, receipts, and administration functions. Here are the 5 client representatives from our company: 1. Nadeesha Perera – Operations Manager 2. Kasun Jayawardena – Senior Driver Coordinator 3. Tharushi Fernando – Customer Experience Officer 4. Ravindu Senanayake – System Administrator 5. Dilini Wickramasinghe – Finance and Payments Manager You can now ask any question about the system requirements. I will respond as the most suitable person from the company.		Next, I want to talk to ms. Tharushi Fernando Copilot Tharushi Fernando (Customer Experience Officer): Hello! Nice to meet you. I work with passengers and handle customer satisfaction. My job is to make sure users have a good experience when using SwiftRide Lanka. Customers can contact us when they have problems such as ride cancellations, driver behavior issues, payment concerns, or account problems. We record complaints, investigate them, and provide solutions. We also collect ratings and feedback after each ride. This helps us improve our service quality and identify drivers who provide excellent service.
okay , i want to talk to ms. Nadeesha Perera Copilot Nadeesha Perera (Operations Manager): Hello! I'm Nadeesha Perera, the Operations Manager at SwiftRide Lanka. Our main goal is to help passengers get a ride quickly and safely. We also want drivers to receive trip requests fairly and manage their daily work easily. The system should be simple to use for both passengers and drivers. Please ask me any questions about the business needs, daily operations, or how the service should work. 🗨️		How do you handle passenger complaints? Copilot Tharushi Fernando (Customer Experience Officer): When a passenger submits a complaint, we first record the complaint with details such as ride ID, date, and the issue reported. Then our customer support team reviews the information. We may contact the passenger and driver to understand what happened. If needed, we also check ride records and payment details. After the investigation, we decide on the appropriate action. This could include providing an explanation, issuing a refund, giving a warning to a driver, or taking other corrective measures. We also keep a history of complaints so we can identify repeated issues and improve our service.
How daily operations works on SwiftRide Lanka? Copilot Nadeesha Perera (Operations Manager): Our daily operations are quite simple. A passenger first creates an account and enters their pickup and drop-off locations. Then they request a ride through the Streetify system. The system sends the request to available drivers nearby. A driver can accept or reject the trip. When a driver accepts, the passenger receives the driver details and vehicle information. After the ride is completed, the passenger makes the payment and receives a receipt. Both the passenger and driver can then leave ratings and feedback.		How do you handle ride cancellations? Copilot Tharushi Fernando (Customer Experience Officer): When a ride is cancelled, the system records who cancelled it, whether it was the passenger or the driver, and the reason for the cancellation. If a passenger cancels, we check whether the cancellation happened before or after a driver accepted the ride. If needed, a cancellation fee may apply according to company policy. If a driver cancels after accepting a ride, we record the reason and monitor the driver's cancellation history. Frequent cancellations may lead to warnings or further review. If a customer complains about a cancellation, our support team investigates the case and helps find a fair solution.

Next, I want to talk to mr.Ravindu Senanayake Copilot Ravindu Senanayake (System Administrator): Hello! Nice to meet you. I am responsible for managing the Streetify system and making sure it runs smoothly every day. I also manage user accounts, monitor system activities, and handle reports related to passengers, drivers, and trips. If there are issues such as fake accounts, policy violations, or complaints that require administrative action, my team reviews them and takes the necessary action.		Tell me about payment methods Copilot Dilini Wickramasinghe (Finance and Payments Manager): At SwiftRide Lanka, we want to provide easy and secure payment options for passengers. Passengers can pay using: • Credit cards • Debit cards • Digital wallets • Cash (if the service allows cash payments)	
What actions do you take for policy violations? Copilot Ravindu Senanayake (System Administrator): Good question. When we find a policy violation, we first review the details and the evidence available in the system. The action depends on how serious the violation is. For minor issues, we may send a warning and educate the user about the company rules. For repeated violations, we may temporarily suspend the account. For serious violations such as fake accounts, fraudulent activities, harassment complaints, or misuse of the platform, we may permanently disable the account after investigation. We also keep records of all administrative actions for future reference and audits.		thank you, i am done for now, i will come back for any further information. Copilot You're welcome! 🌟 It was nice helping you gather requirements for the Streetify Web-Based Transport System.	
finally, I want to talk to Ms. Dilini Wickramasinghe Copilot Dilini Wickramasinghe (Finance and Payments Manager): Hello! Nice to meet you. I manage all payment-related activities at SwiftRide Lanka. My job is to make sure payments are processed correctly and that both passengers and drivers receive accurate financial records. You can ask me about payments, receipts, fare calculations, refunds, driver earnings, or financial reports. 🌟		So far, you have collected useful information about: • Daily operations • Driver trip acceptance and cancellation • Passenger complaints and ride cancellations • User account management • Policy violations • Fare calculation • Payment methods • Financial reporting This information should help you identify functional requirements, business rules, and user needs for your Software Engineering module. Whenever you need more information, just come back and continue the discussion. You can talk to any of the SwiftRide Lanka representatives again, and I will respond as that person.	

Phase 1: Domain Exploration & Role Clarification

Hi, I'm doing a software project for our software engineering module. Can you help with that? Copilot Of course! I'd be happy to help with your software engineering project. First, please tell me a little about your project: 1. What is the project title? 2. Who will use the system? 3. Is it a web application, mobile app, desktop system, or something else? 4. What does the system do? Once you give these details, I will create a Sri Lankan client company scenario with 3 different people (personas) from that company. Then you can talk to them and collect requirements just like a real client meeting. 🌟		1.Steetify web based transport system like uber 2.passengers,drivers,system administrators 3.web application 4.user account management ,ride booking,driver trip request management,review and feedback management,payment and receipt system,admin and system control
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Phase 2: Use Cases & Exception Flows

No-show fee application & ride cancellations How do drivers accept or cancel trips? Copilot Kasun Jayawardena (Senior Driver Coordinator): Sure. When a new ride request comes, available drivers receive a notification. They can see basic trip details such as pickup location and drop-off location. If a driver wants the trip, they click Accept Ride. After accepting, the system sends the driver's details to the passenger. Sometimes drivers need to cancel a trip. This may happen because of a vehicle problem, an emergency, or difficulty reaching the pickup location. When cancelling, the driver should give a reason. We monitor cancellations because too many cancellations can affect service quality. For our business, it is important that drivers do not cancel rides often after accepting them.	Payment retry loops & gateway failures	Next, I want to talk to ms. Tharushi Fernando Copilot Tharushi Fernando (Customer Experience Officer): Hello! Nice to meet you. I work with passengers and handle customer satisfaction. My job is to make sure users have a good experience when using SwiftRide Lanka. Customers can contact us when they have problems such as ride cancellations, driver behavior issues, payment concerns, or account problems. We record complaints, investigate them, and provide solutions. We also collect ratings and feedback after each ride. This helps us improve our service quality and identify drivers who provide excellent service.
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Phase 3: Database & Attribute Modeling

finally, I want to talk to Ms. Dilini Wickramasinghe Copilot Dilini Wickramasinghe (Finance and Payments Manager): Hello! Nice to meet you. I manage all payment-related activities at SwiftRide Lanka. My job is to make sure payments are processed correctly and that both passengers and drivers receive accurate financial records. You can ask me about payments, receipts, fare calculations, refunds, driver earnings, or financial reports. 🌟	thank you, i am done for now, i will come back for any further information. Copilot You're welcome! 🌟 It was nice helping you gather requirements for the Streetify Web-Based Transport System.	So far, you have collected useful information about: • Daily operations • Driver trip acceptance and cancellation • Passenger complaints and ride cancellations • User account management • Policy violations • Fare calculation • Payment methods • Financial reporting This information should help you identify functional requirements, business rules, and user needs for your Software Engineering module. Whenever you need more information, just come back and continue the discussion. You can talk to any of the SwiftRide Lanka representatives again, and I will respond as that person.
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